

VoIP Overview

VoIP – Voice over IP

PSTN – Public Switched Telephone Network (Štandardná telefónna sieť)

How voice data are created?

1. Hlas naráža do membrány v mikrofóne, tak sa mení elektrický prúd a vytvárajú sa vlnovky
2. Dané vlnovky preženieme cez Codec G.711 (Vzorkovanie -> Kvantovanie -> Kódovanie) -> po tomto z vlnoviek budú 1 a 0 [Codec G.711 = Sampling, Quantizing, Encoding]
 - a. Každý device podporuje 64kbit/s

Codecs

1. G.711
 - a. Vzorkovanie -> Kvantovanie -> Kódovanie (Sampling, Quantizing, Encoding)
 - b. Rychlosť = 64kbit/s
 - c. Typicky sa používa na interných LANkách [ma väčšiu bandwidth, na LANke nam to je jedno]
2. G.729
 - a. Rychlosť = 8kbit/s
 - b. Typicky sa používa vo WANKách [lebo má malú bandwidth]

RTP – Real Time Transport Protocol

- Protokol Aplikačnej vrstvy
- Prenáša Audio/Video
- Využíva QoS

Čo robí RTP?

1. Má synchronizačné vlny a buffer
2. Pozrie sa kedy prišiel ktorý packet a podľa neho si do bufferu ukladá že čo prišlo ako ktoré

QoS – Quality of Service

- Markuje dáta, dáva ich do tried a podľa tých tried sa prioritizujú packety

Čo potrebujem na VOIP sieť ?

1. Call Server [Telefón]
2. Call processor [v našom prípade Call Manager]
3. Gateway [Router s nejakou voice kartou]

SIP – Session Initiation Protocol

- Najviac používaný signalizačný protokol

SCCP = Skinny Client xxx

SIP Komunikácia

- Podobné http
- Client-Server protokol
- SIP dialóg = SIP Offer (Invite, act, bye) + SIP response

RTP Stream

- separátna komunikácia, nezávislá na SIP, môže ísť inou cestou
- SIP Invite -> Ringing -> Acknowledge -> Act (uz nejde cez call manager) -> Bye

Musíš vedieť:

Vzorkovanie -> Kvantovanie -> Kódovanie (Zaradom a vysvetliť)

Codec G.711 (Čo to je + rýchlosť)

Codec G.729 (Čo to je + rýchlosť)

Simple info o RTP

Čo potrebujem na VOIP sieť

Vyber signalizačný protokol (SIP)

SIP Komunikácia + RTP Stream

Slide 24 – Signalling Protocols

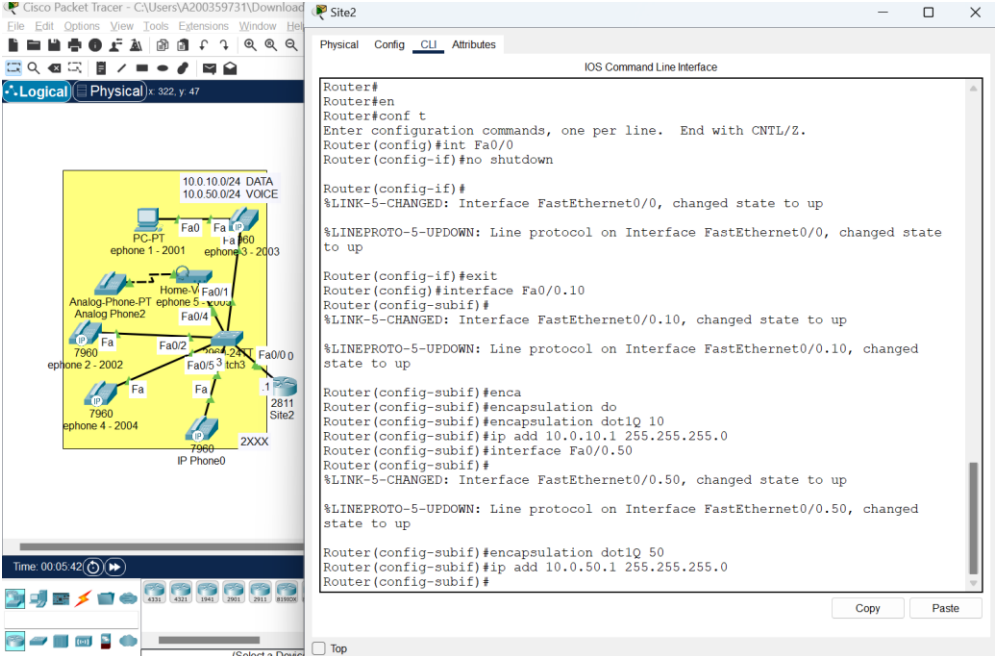
24.11

Phone bot process

1. Telefon v sieti
2. Naladuje sa
3. Vyhľadá pomocou CDP (Cisco Discovery Protocol) susedov (zariadenia naokolo)
4. Vyžiada IP od DHCP servera
5. Telefón sa spojí s DHCP
6. Telefón dostane IP, masku, default gateway

Konfiguracia:

1. Treba rozbehať router
 - a. Konfigurácia interfacov (Router on a stick -> ideme robiť VLANy)



The screenshot displays the Cisco Packet Tracer interface. On the left, a network diagram shows a central router (Site2) with multiple interfaces connected to various devices including PCs, analog phones, and IP phones. The configuration window on the right shows the CLI commands for configuring the router interfaces and subinterfaces.

```
Router#
Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int Fa0/0
Router(config-if)#no shutdown

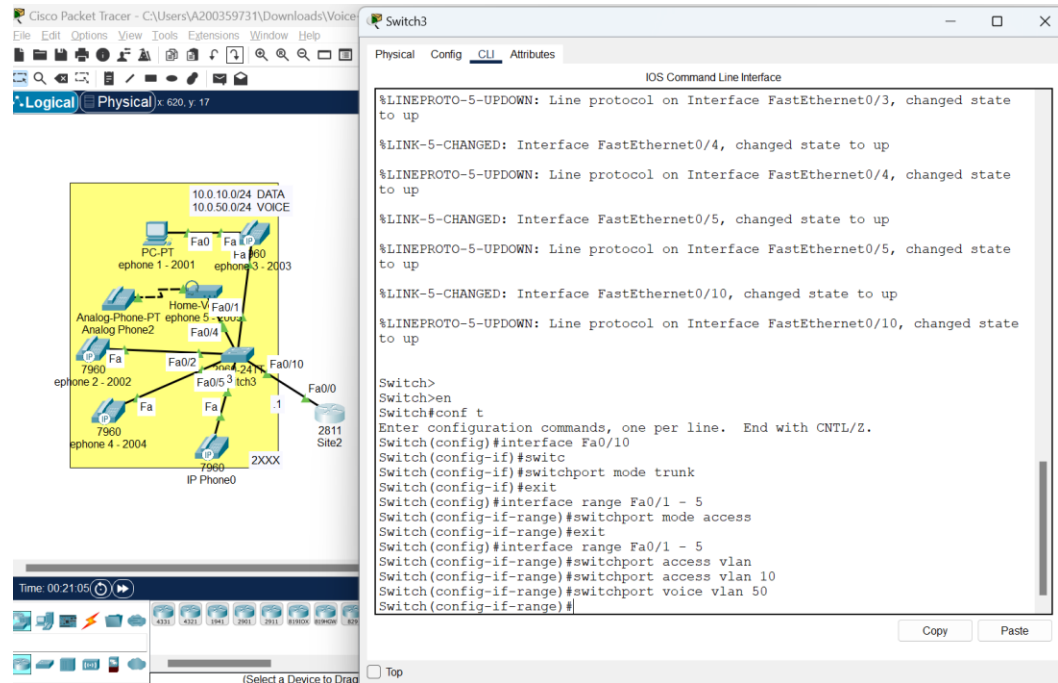
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Fa0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Router(config-subif)#enca
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 10.0.10.1 255.255.255.0
Router(config-subif)#interface Fa0/0.50
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.50, changed state to up

Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip add 10.0.50.1 255.255.255.0
Router(config-subif)#
```

b. Nastavíme SWTICH (Trunk) [trunk povolí aj Vlan10 aj Vlan50]



The screenshot shows the Cisco Packet Tracer interface. On the left, a network diagram is visible with various devices including PCs, phones, and switches. On the right, the CLI of a switch named 'Switch3' is open, showing the configuration process for a trunk port.

```

Switch3
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up

Switch>
Switch>en
Switch>conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface Fa0/10
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#interface range Fa0/1 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#exit
Switch(config)#interface range Fa0/1 - 5
Switch(config-if-range)#switchport access vlan
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#switchport voice vlan 50
Switch(config-if-range)#
  
```

c. (OPTION 150 [TFTP] ip [TFTP server -> u nas router lebo router on a stick])

```

Router(config)#ip dh
Router(config)#ip dhcp po
Router(config)#ip dhcp pool DATA
Router(dhcp-config)#network 10.0.1.0 255.255.255.0
Router(dhcp-config)#default-router 10.0.10.1
Router(dhcp-config)#e
Router(config)#ip d
Router(config)#ip dhcp pool VOICE
Router(dhcp-config)#network 10.0.50.0 255.255.255.0
Router(dhcp-config)#option 150 ip 10.0.50.1

% Invalid input detected at '^' marker.

Router(dhcp-config)#network 10.0.50.0 255.255.255.0
Router(dhcp-config)#default-router 10.0.5.1
Router(dhcp-config)#option 150 ip 10.0.50.1
  
```

d. Telephony service

```

Router(config)#telephony-service
Router(config-telephony)#max-dn 144
Router(config-telephony)#max-ephones 42
Router(config-telephony)#ip source-address 10.0.50.1 port 2000
  
```

e. Ephones

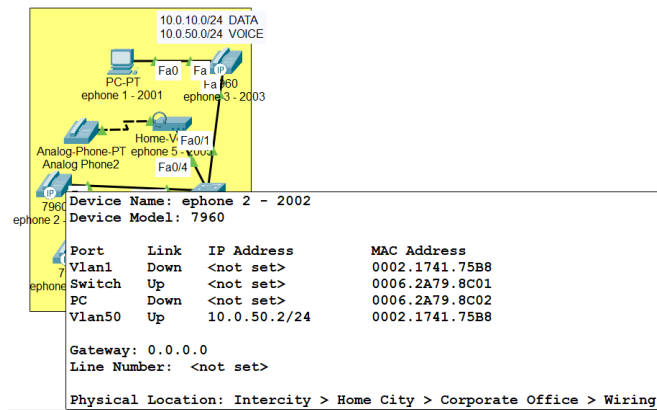
```
Router(config)#  
Router(config)#ephone-dn 1  
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed  
state to up  
  
Router(config-ephone-dn)#number 2001  
Router(config-ephone-dn)#exit  
Router(config)#ephone-dn 2  
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed  
state to up  
  
Router(config-ephone-dn)#number 2002  
Router(config-ephone-dn)#exit  
Router(config)#ephone-dn 3  
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 3.1, changed  
state to up  
  
Router(config-ephone-dn)#number 2003  
Router(config-ephone-dn)#exit  
Router(config)#ephone-dn 4  
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 4.1, changed  
state to up  
  
Router(config-ephone-dn)#number 2004  
Router(config-ephone-dn)#  
Router(config-ephone-dn)#exit  
Router(config)#  
Router(config)#ephone-dn 5  
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 5.1, changed  
state to up  
  
Router(config-ephone-dn)#number 2005
```

DO SH RUN

```
!  
telephony-service  
max-ephones 42  
max-dn 144  
ip source-address 10.0.50.1 port 2000  
!  
ephone-dn 1  
number 2001  
!  
ephone-dn 2  
number 2002  
!  
ephone-dn 3  
number 2003  
!  
ephone-dn 4  
number 2004  
!  
ephone-dn 5  
number 2005  
!
```

Dame write a reload

Zabudli sme na toto :
Pridame veci a mac
(MAC TUTO Z VLAN 50)



COMMANDY CO SME ZABUDLI + MAC :

```
Router(config)#telephony-service
Router(config-telephony)#auto-reg-ephone
Router(config-telephony)#exit
Router(config)#ephone 2
Router(config-ephone)#mac
Router(config-ephone)#mac-address 0002.1741.75B8
```

f. Pridelíme čísla (na gombík 1 dáme číslo 1)

```
Router(config)#ephone 2
Router(config-ephone)#button 1:1
```

g. Reset packet tracera a kukni tu, a ked tu je cislo tka gut



ZVONII !!!!!!!!

